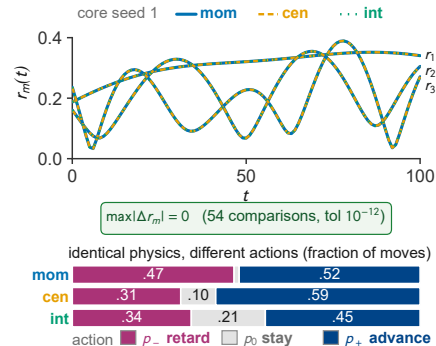
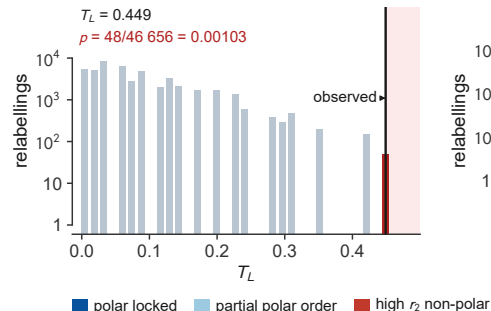


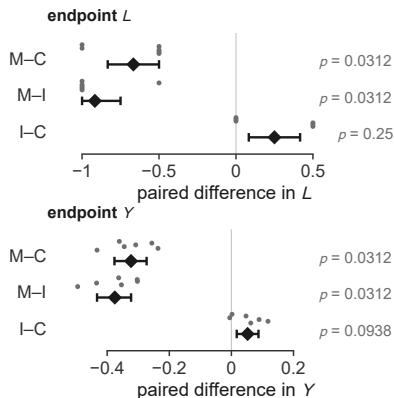
a Exact $K=0$ engine control



d Exact null for T_L

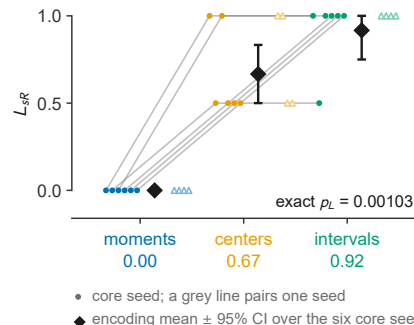


g Pairwise seed-level contrasts

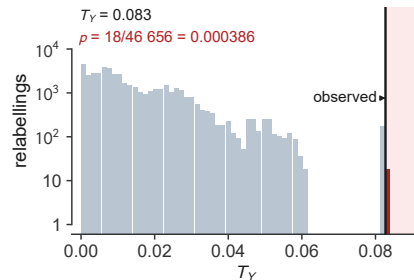


Grey dots: the six core-seed differences.
Diamond: mean with the stored bootstrap 95% CI.
 p : exact sign-flip test over all 64 sign patterns.

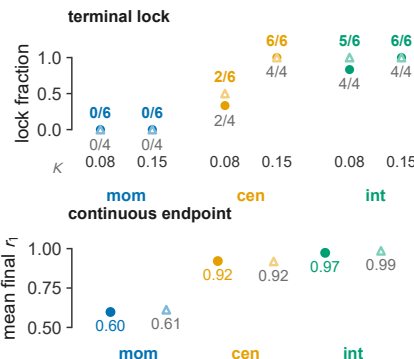
b Terminal-lock score L



e Exact null for T_Y

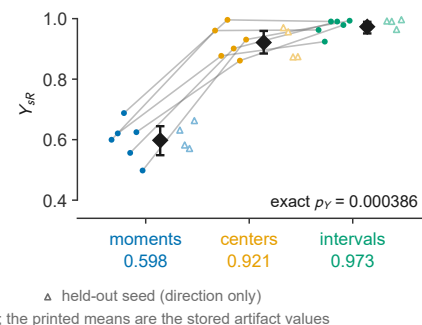


h Core vs held-out direction

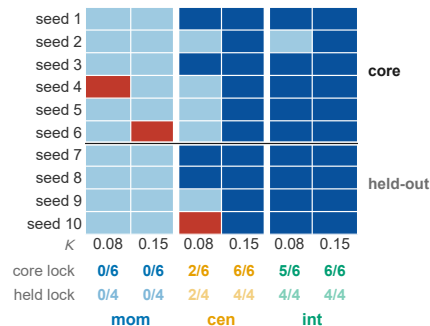


Filled circle: core six. Open triangle: held-out four.
Counts above each column are core, below are held-out.

c Final polar order γ



f Phenotype matrix



i Prespecified gate verdicts

PASS $K=0$ control

engine and matching valid
 $\max|\Delta r_m| = 0$, tol 10^{-12}

PASS Gate A

macro encoding effect
 $p_L = 0.00103$, $p_Y = 0.000386$

PASS Gate B

qualitative phase separation
lock fraction at $K = 0.08$: 0.00, 0.33, 0.83

FAIL Gate C

GPT phenotype map not replicated
final r_1 reverses: $0.60 < 0.92 < 0.97$